

Visualization Report

This report presents an exploratory and business focused analysis of hotel booking data to understand customer booking behaviour, cancellation trends, channel performance, room preferences and profitability drivers. The analysis examines key factors such as booking channels, room types, star ratings, booking values, stay lengths, lead times, refunds and promotional activities.

The objective is to identify meaningful patterns, uncover potential causes behind booking cancellations and revenue variations, and provide actionable recommendations that can help improve operational efficiency, customer retention and overall business profitability. The findings are supported through data visualizations and statistical analysis to enable data-driven decision-making.

The insights and recommendations presented in this report are based solely on the available dataset and the quality of information provided. Results may be influenced by data limitations, missing values, business rules not captured within the dataset or external factors such as market conditions, seasonality, economic changes, and customer preferences. Therefore, the findings should be considered directional and used alongside domain expertise and business judgment when making strategic decisions.

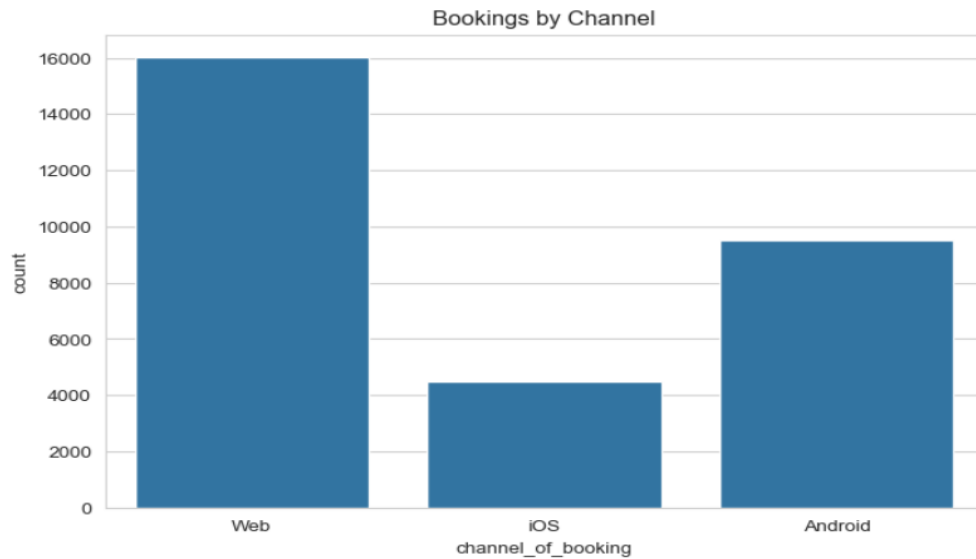
1. Overall Booking Status



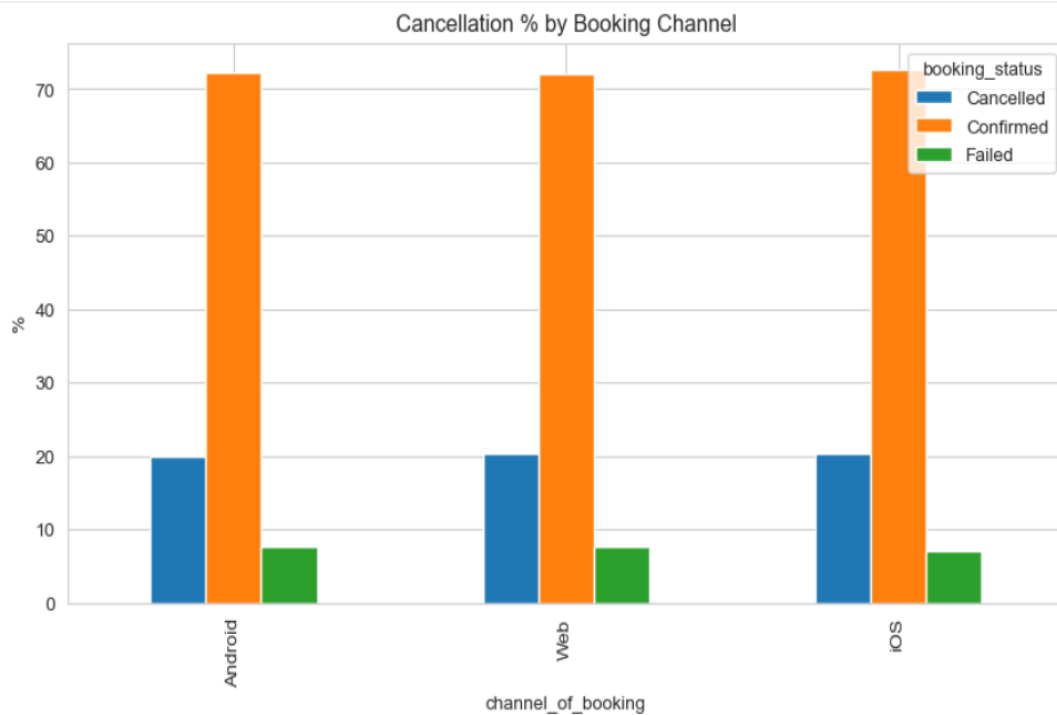
```
booking_status
Confirmed    21672
Cancelled    6070
Failed       2258
Name: count, dtype: int64
Cancellation Rate: 20.23%
```

2. Booking Channel Analysis

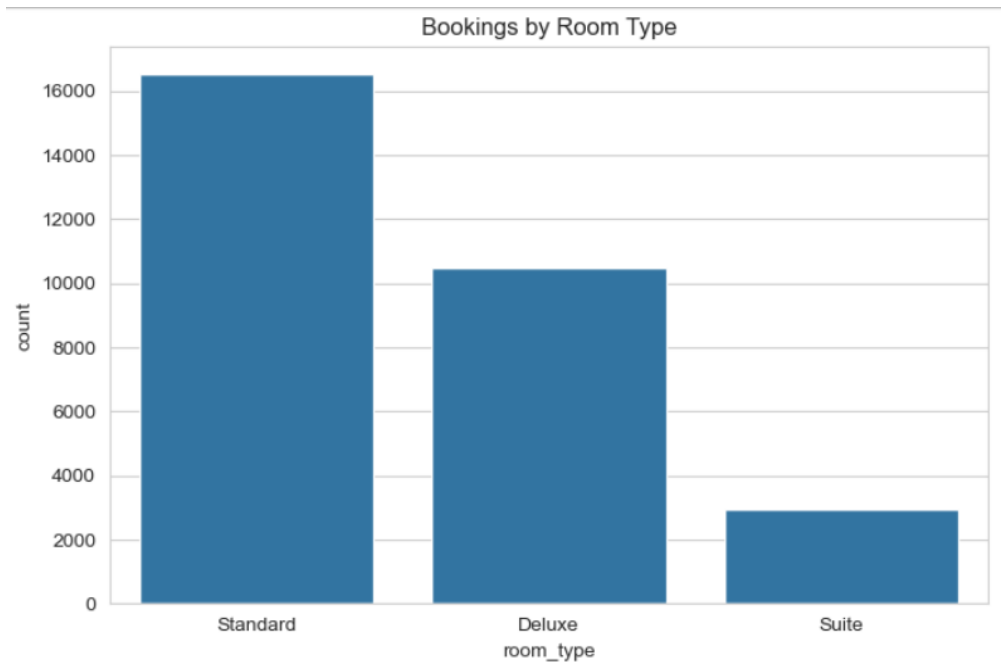
	channel_of_booking	total_bookings	avg_booking_value
0	Android	9505	25355.242358
1	Web	16023	24973.016853
2	iOS	4472	24881.828258



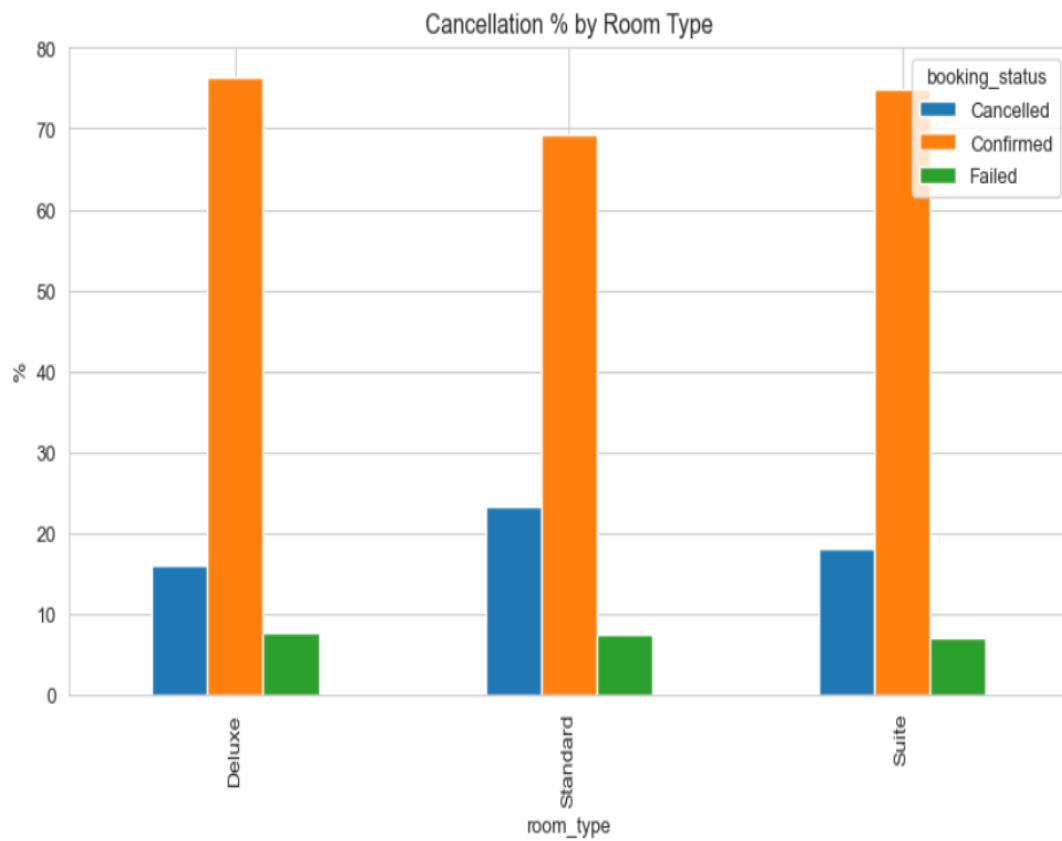
	Cancelled	Confirmed	Failed
channel_of_booking			
Android	20.000000	72.330352	7.669648
Web	20.333271	72.077638	7.589091
iOS	20.371199	72.629696	6.999106



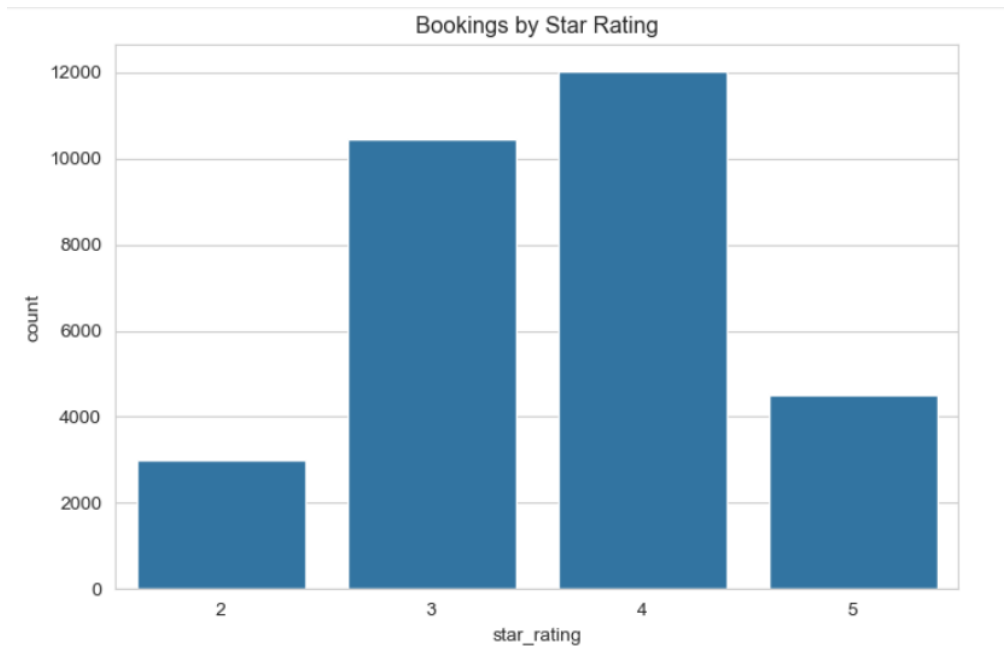
3. Room Type Analysis



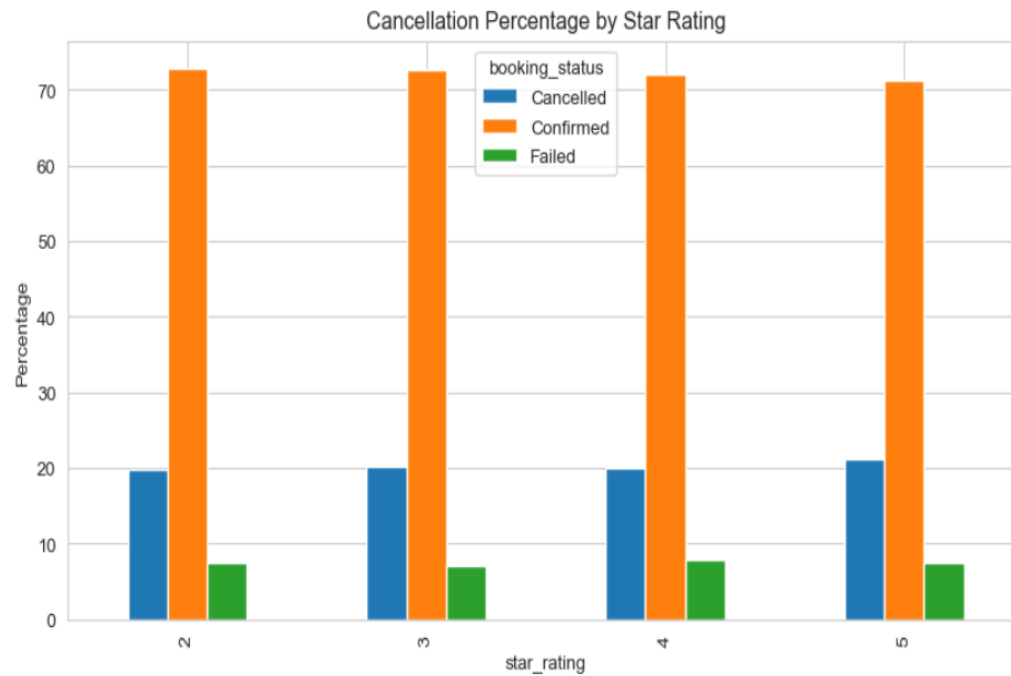
booking_status	Cancelled	Confirmed	Failed
room_type			
Deluxe	16.024050	76.312273	7.663676
Standard	23.302320	69.188014	7.509667
Suite	17.979798	74.882155	7.138047



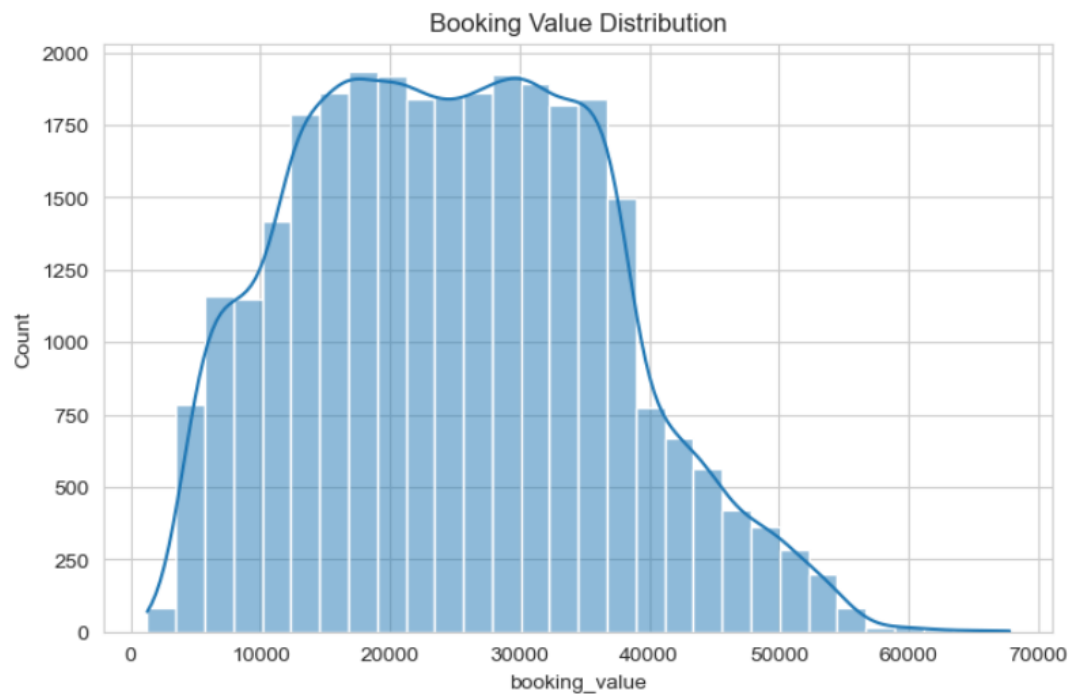
4. Star Rating Analysis



booking_status	Cancelled	Confirmed	Failed
star_rating			
2	19.766277	72.854758	7.378965
3	20.200765	72.648184	7.151052
4	19.993352	72.112348	7.894299
5	21.259144	71.225892	7.514963



5. Booking Value Analysis

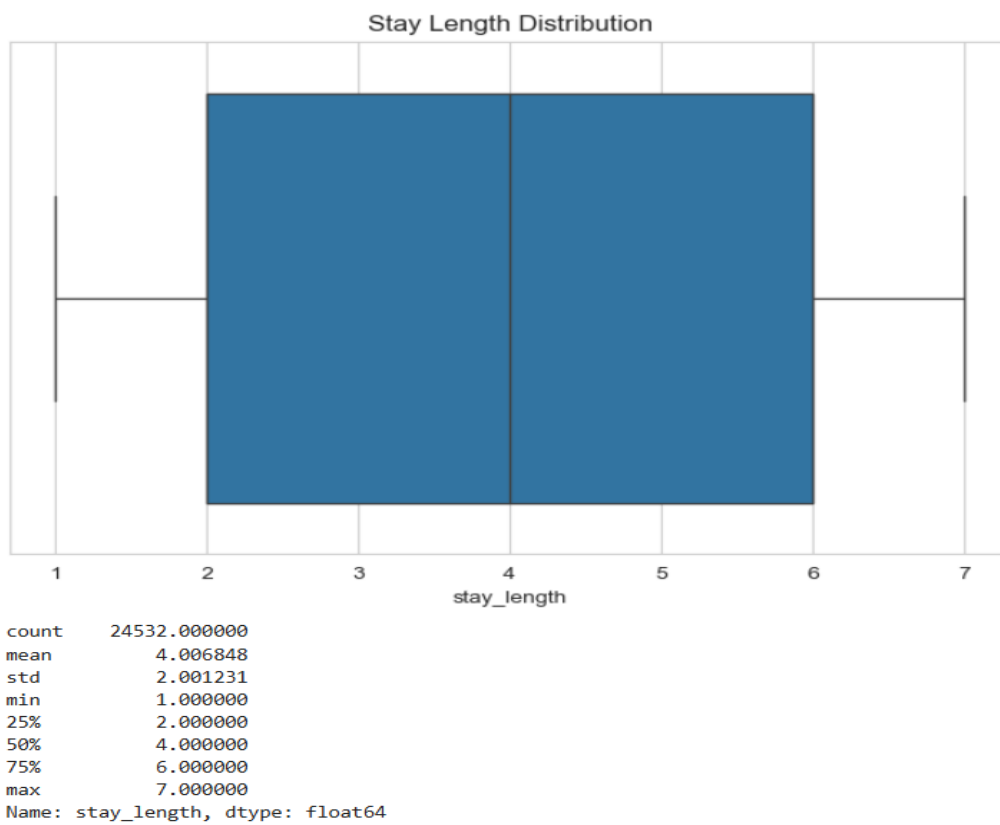


```
count    30000.000000
mean     25080.525454
std      11587.369335
min       1279.770000
25%      15922.712500
50%      24736.145000
75%      33570.000000
max       67764.460000
Name: booking_value, dtype: float64
```

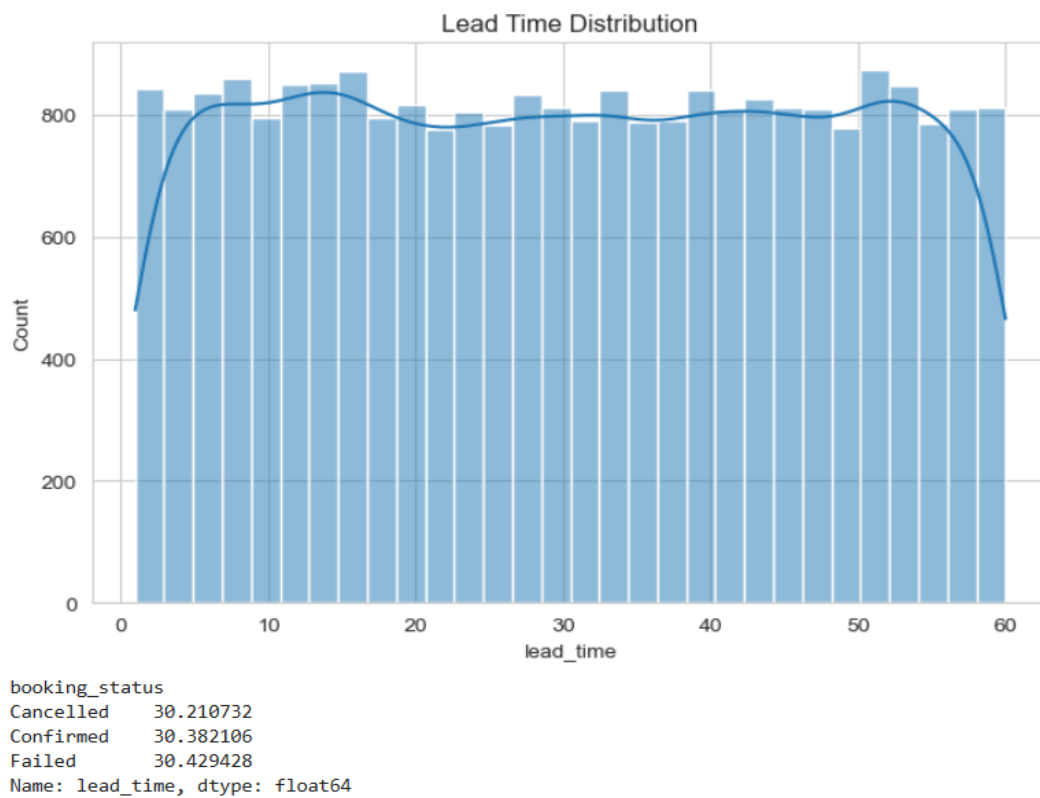
6. Profitability Analysis



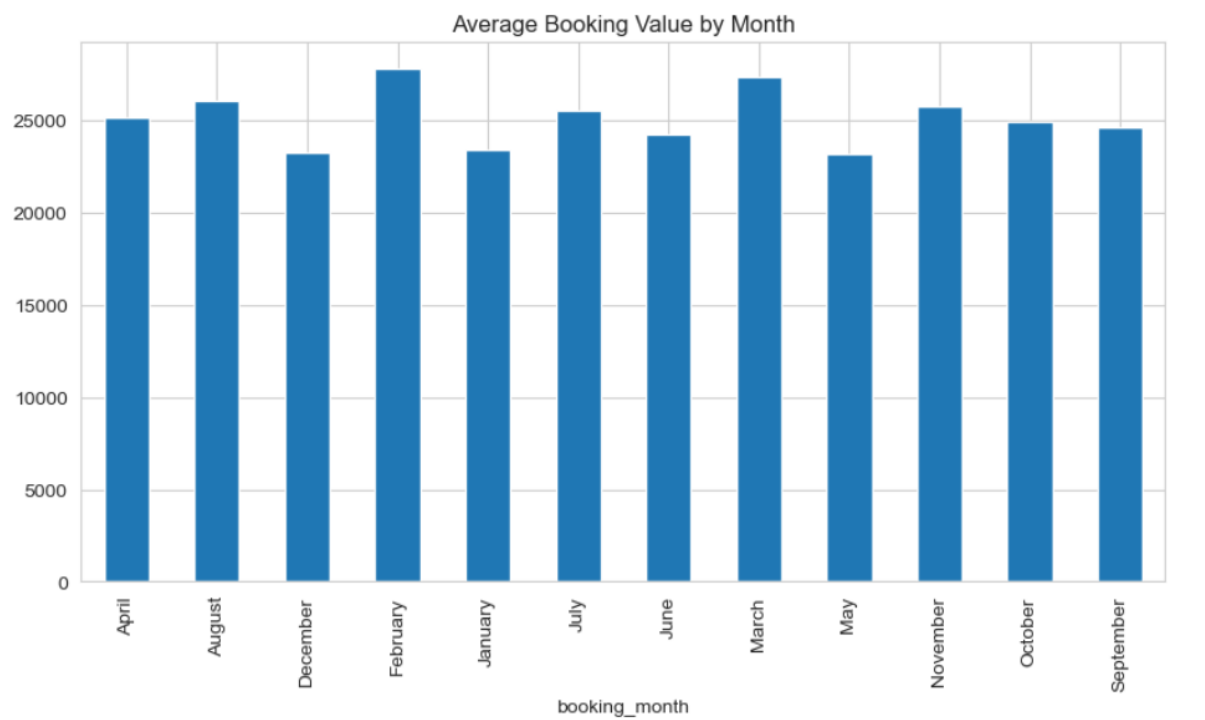
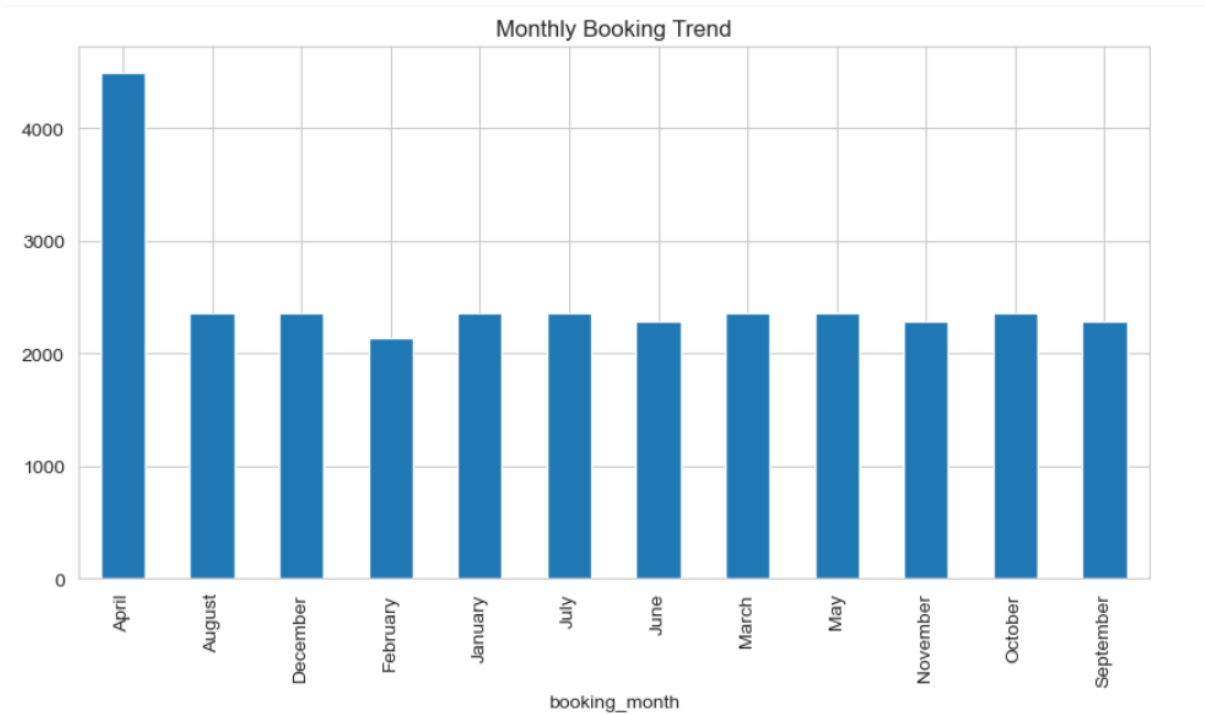
7. Stay Length Analysis



8. Lead Time Analysis



9. Monthly Trend Analysis



10. Top Revenue Generators

